Ethan Yap

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SUMMARY

IT student seeking a software engineering internship. Builds and deploys full-stack web apps in Node.js, Next.js and Python on PostgreSQL, from REST APIs to AI search, with Docker and GitHub Actions CI/CD.

EDUCATION

Singapore Polytechnic · Diploma in IT (Software and Applications), Minor in Business — Apr 2025 – 2028 (expected)
Cumulative GPA 3.43 / 4.00 · Key modules: Back-End Web Development, Java Programming, Continuous Integration and Continuous Delivery, Fundamentals of Programming, Design for User Interaction

Maris Stella High School · GCE O-Level, L1R5 net 10 · Direct School Admission (Robotics) — 2021 – 2024

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Java, SQL, HTML, CSS
Backend	Node.js, Express, FastAPI, Spring Boot, REST APIs, WebSockets, JWT auth, background job queues
Data & AI	PostgreSQL, pgvector, MySQL, SQLite, pandas, ETL, embeddings, RAG, ChromaDB, Gemini API
Frontend	React, Next.js, Tailwind CSS, Bootstrap, Chart.js, Recharts, responsive and accessible UI, Java Swing
DevOps & testing	Docker, Docker Compose, nginx, GitHub Actions (CI/CD), Vercel, Neon, Git, pytest, node:test, Zod, Figma
Spoken	English, Mandarin

PROJECTS

Incident Monitoring Dashboard · *Docker Compose, Node.js, PostgreSQL, WebSockets, Next.js* — Code · Demo
- Designed a 5-container Docker Compose stack with health checks, verified end to end by a smoke test in GitHub Actions.
- Built an alerting engine whose hysteresis (2 bad readings to open, 3 good to resolve) filters noisy spikes; 15 unit tests.

AI Code Snippet & Documentation Hub · *Next.js, TypeScript, PostgreSQL, pgvector, Gemini API* — Code · Live · Demo
- Kept saves instant with a background worker on a Postgres SKIP LOCKED job queue that writes Zod-validated LLM docs.
- Built semantic search over 768-dimension embeddings with a pgvector HNSW index, re-ranked by full-text score; 73 tests.

Academic Resource RAG Search Engine · *Python, FastAPI, ChromaDB, sentence embeddings, pytest* — Code · Demo
- Built PDF ingestion that packs sentence-aligned chunks by page and section, so every answer cites its exact page.
- Refused off-topic questions with a relevance cut-off and rejected any uncited LLM answer, covered by 26 pytest tests in CI.

Tech Trend Sentiment Tracker · *Python, pandas, SQLite, FastAPI, Next.js, Recharts* — Code · Demo
- Automated a 6-hourly GitHub Actions ETL pipeline that de-duplicates news from 8 sources and scores its sentiment.
- Kept the Python API and the JavaScript dashboard in agreement with a parity test, one of 18 pytest tests run in CI.

Pokémon Gym & Battle System · *Node.js, Express, PostgreSQL, JWT, bcrypt, node:test* — Code · Live · Demo
- Built a 45-endpoint REST API with JWT auth and ownership checks, then ported it from MySQL to PostgreSQL.
- Wrote a battle engine and a transactional gacha shop, covered by 20 API tests run in CI against a real PostgreSQL.

Steady: Weight & Calorie Tracker · *Node.js, Express, PostgreSQL, JWT, bcrypt, node:test* — Code · Live · Demo
- Wrote a parser turning “half a plate of chicken rice” into grams across 242 foods; 140 tests run in CI on real Postgres.

Slotted: Facility Booking System (in progress) · *Spring Boot, Java 21, PostgreSQL, React* — Code
- Built conflict-free booking with a Postgres exclusion constraint and concurrency tests, among 184 API tests run in CI.

MyLearningAI Study App · *JavaScript, HTML, CSS, Figma (group of 4)* — Code · Demo
- Built our team’s Figma design into a 60+ screen app; usability tests added a progress/time indicator and flag/skip.

WORK EXPERIENCE

Two Men Bagel House · Line Chef (part-time) — Feb 2025 – Apr 2026
- Coordinated orders between kitchen and front of house at peak service; ran the register and barista station.

Bee Cheng Hiang · Barbecue Cook (part-time) — Nov 2024 – Feb 2025
- Ran a cooking station alone through the festive peak, sequencing orders ahead of demand to shorten queues.

LEADERSHIP, COMMUNITY & AWARDS

Marist Ambassador · frequent emcee at school events incl. prize-giving; volunteered at multiple Open Houses — 2021 – 2024

Marist Robotics & Programming Club · mentored juniors; Perse Coding Team Challenge 2022–23 (**Merit Award**) — 2021 – 2024

Community service · ran registration for Mummy Yummy events of 100+ people; Magorium, Masjid Kassim — 2022 – 2025

Edusave Character Award (MOE and People’s Association) · **Class Chairman** (Sec 4) · **Peer Supporter** (Sec 2) — 2022 – 2024